

Assignment 9: Spatial Analysis in R

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics (ENV872L) on spatial analysis.

Directions

1. Rename this file `<FirstLast>_A09_SpatialAnalysis.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Use the lesson as a guide. It contains code that can be modified to complete the assignment.
4. Work through the steps, **creating code and output** that fulfill each instruction.
5. Be sure to **answer the questions** in this assignment document. Space for your answers is provided in this document and is indicated by the “>” character. If you need a second paragraph be sure to start the first line with “>”. You should notice that the answer is highlighted in green by RStudio.
6. When you have completed the assignment, **Knit** the text and code into a single **HTML** file.

DATA WRANGLING

Set up your session

1. Import libraries: tidyverse, sf, leaflet, here, and mapview
2. Execute the `here()` command to display the current project directory

#1.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(leaflet)
library(here)
```

```
## here() starts at /Users/kaitlynkukula/EDE_Fall2025
```

```
library(mapview)
library(lubridate)
```

```
#2.
here()
```

```
## [1] "/Users/kaitlynkukula/EDE_Fall2025"
```

```
# Set ggplot theme
kak_theme <- theme_classic() +
  theme(plot.title = element_text(size = 11, hjust = 0.5, face = "bold"), #Adjust title
        axis.text.x = element_text(size = 9), # Adjust x-axis values
        axis.text.y = element_text(size = 9), # Adjust y-axis values
        axis.title.x = element_text(size = 10.5, face = "bold"), # Adjust x-axis title
        axis.title.y = element_text(size = 10.5, face = "bold"), # Adjust y-axis title
        legend.position = "right", # Define legend position
        legend.text = element_text(size = 9), # Define legend entry sizes
        legend.title = element_text(size = 9), # Define legend name sizes
        plot.subtitle = element_text(hjust = 0.5))
set_theme(kak_theme)
```

Read (and filter) county features into an sf dataframe and plot

In this exercise, we will be exploring stream gage height data in Nebraska corresponding to floods occurring there in 2019. First, we will import from the US Counties shapefile we've used in lab lessons, filtering it this time for just Nebraska counties. Nebraska's state FIPS code is 31 (as North Carolina's was 37).

3. Read the `cb_2018_us_county_20m.shp` shapefile into an sf dataframe, filtering records for Nebraska counties (State FIPS = 31)
4. Reveal the dataset's coordinate reference system
5. Plot the records as a map (using `mapview` or `ggplot`)

```
#3. Read in Counties shapefile into an sf dataframe, filtering for just NE counties
```

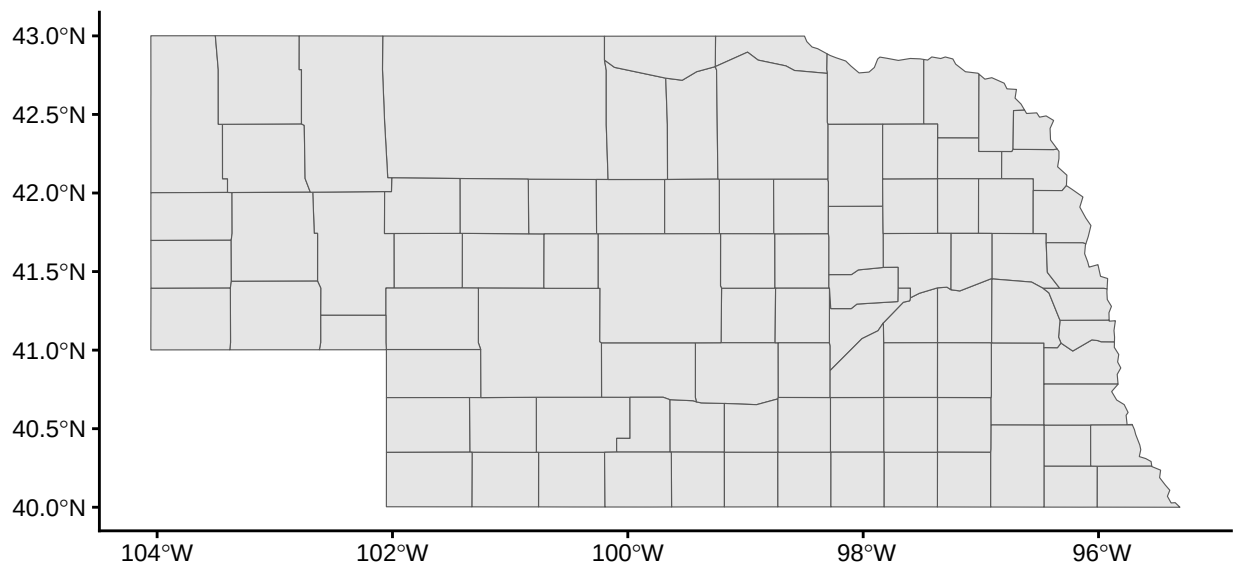
```
nebraska_counties <- st_read(here('Data/Spatial/cb_2018_us_county_20m.shp'),
                             stringsAsFactors = TRUE) %>%
  filter(STATEFP == "31")
```

```
## Reading layer 'cb_2018_us_county_20m' from data source
##   '/Users/kaitlynkukula/EDE_Fall2025/Data/Spatial/cb_2018_us_county_20m.shp'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 3220 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -179.1743 ymin: 17.91377 xmax: 179.7739 ymax: 71.35256
## Geodetic CRS:   NAD83
```

```
#4. Reveal the CRS of the counties features
st_crs(nebraska_counties) # NAD83
```

```
## Coordinate Reference System:
##   User input: NAD83
##   wkt:
##   GEOGCRS["NAD83",
##     DATUM["North American Datum 1983",
##       ELLIPSOID["GRS 1980",6378137,298.257222101,
##         LENGTHUNIT["metre",1]],
##     PRIMEM["Greenwich",0,
##       ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##       AXIS["latitude",north,
##         ORDER[1],
##         ANGLEUNIT["degree",0.0174532925199433]],
##       AXIS["longitude",east,
##         ORDER[2],
##         ANGLEUNIT["degree",0.0174532925199433]],
##     ID["EPSG",4269]]
```

```
#5. Plot the data
ggplot(nebraska_counties) + geom_sf()
```



6. What is the EPSG code of the Counties dataset? Is this a geographic or a projected coordinate reference

system? (In other words, does this CRS use angular or planar coordinate units?) To what datum is this CRS associated? (Tip: lookup the EPSG code on <https://epsg.io> or <https://spatialreference.org>)

ANSWER: The EPSG code is 4269. This corresponds to NAD83, which is a geographic coordinate system. The datum associated with NAD83 is the North American Datum developed in 1983 (ellipsoid is GRS1980) where the Prime meridian passes through Greenwich, UK.

Read in gage locations csv as a dataframe, then display the column names it contains

Next we'll read in some USGS/NWIS gage location data added to the `Data/Raw` folder. These are in the `NWIS_SiteInfo_NE_RAW.csv` file. (See `NWIS_SiteInfo_NE_RAW.README.txt` for more info on this dataset.)

7. Read the `NWIS_SiteInfo_NE_RAW.csv` file into a standard dataframe, being sure to set the `site_no` field as well as other character columns as a factor.
8. Display the structure of this dataframe.

```
#7. Read in gage locations csv as a dataframe
nebraska_gages <- read.csv('Data/Raw/NWIS_SiteInfo_NE_RAW.csv',
                           stringsAsFactors = TRUE)

#8. Display the structure of the dataframe
glimpse(nebraska_gages)
```

```
## Rows: 175
## Columns: 7
## $ site_no      <int> 6453600, 6454100, 6461500, 6463500, 6463720, 646550~
## $ station_nm   <fct> "Ponca Creek at Verdel, Nebr.", "Niobrara River at ~
## $ site_tp_cd    <fct> ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ~
## $ dec_lat_va    <dbl> 42.81111, 42.42361, 42.90222, 42.68944, 42.78056, 4~
## $ dec_long_va   <dbl> -98.17583, -103.79222, -100.36222, -99.67889, -99.3~
## $ coord_acy_cd  <fct> S, S, S, S, S, S, S, S, S, S, 1, F, S, S, F, H, S, ~
## $ dec_coord_datum_cd <fct> NAD83, NAD83, NAD83, NAD83, NAD83, NAD83, NAD83, NA~
```

9. What columns in the dataset contain the x and y coordinate values, respectively?
> ANSWER: `dec_lat_va` contains the latitude (y) coordinates, and `dec_long_va` contains the longitude coordinates (x).

Convert the dataframe to a spatial features ("sf") dataframe

10. Convert the dataframe to an sf dataframe. * Note: These data use the same coordinate reference system as the counties dataset
11. Display the structure of the resulting sf dataframe

```
#10. Convert to an sf object
nebraska_gages_sf <- nebraska_gages %>%
  st_as_sf(coords = c("dec_long_va", "dec_lat_va"),
           crs = 4269)

#11. Display the structure
glimpse(nebraska_gages_sf)
```

```
## Rows: 175
## Columns: 6
## $ site_no      <int> 6453600, 6454100, 6461500, 6463500, 6463720, 646550~
## $ station_nm   <fct> "Ponca Creek at Verdel, Nebr.", "Niobrara River at ~
## $ site_tp_cd   <fct> ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ST, ~
## $ coord_acy_cd <fct> S, S, S, S, S, S, S, S, S, S, S, 1, F, S, S, F, H, S, ~
## $ dec_coord_datum_cd <fct> NAD83, NAD83, NAD83, NAD83, NAD83, NAD83, NAD83, NA~
## $ geometry     <POINT [°]> POINT (-98.17583 42.81111), POINT (-103.7922 ~
```

12. What new field(s) appear in the sf dataframe created? What field(s), if any, disappeared?

ANSWER: The fields for latitude and longitude disappeared but were used to generate a new field containing geometry.

Plot the gage locations on top of the counties

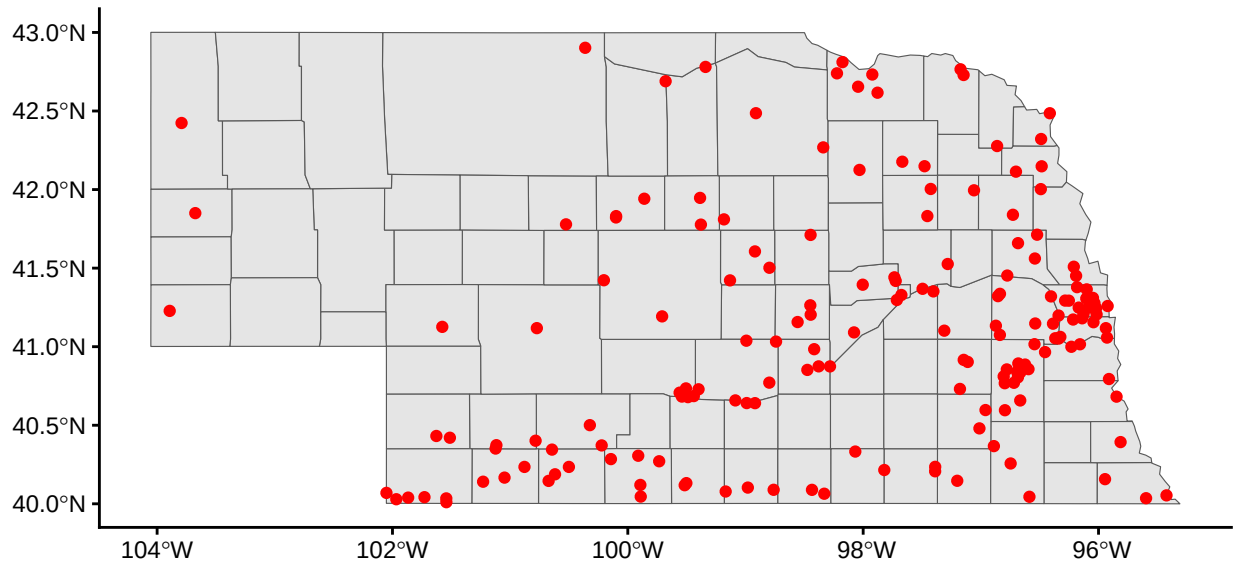
13. Use `ggplot` to plot the county and gage location datasets.

- Be sure the datasets are displayed in different colors
- Title your plot “NWIS Gage Locations in Nebraska”
- Subtitle your plot with your name

```
#13. Plot the gage locations atop the county features
ggplot() +
  geom_sf(data = nebraska_counties) +
  geom_sf(data = nebraska_gages_sf, color = 'red') +
  labs(title = "NWIS Gage Locations in Nebraska",
       subtitle = 'Kaitlyn Kukuka')
```

NWIS Gage Locations in Nebraska

Kaitlyn Kukuka



Read in the gage height data and join the site location data to it.

Lastly, we want to attach some gage height data to our site locations. I've constructed a csv file listing many of the Nebraska gage sites, by station name and site number along with stream gage heights (in meters) recorded during the recent flood event. This file is titled `NWIS_SiteFlowData_NE_RAW.csv` and is found in the `Data/Raw` folder.

14. Read the `NWIS_SiteFlowData_NE_RAW.csv` dataset in as a dataframe

- Pay attention to which fields should be imported as factors!

15. Show the structure of the dataframe.

16. Join our site information (already imported above) to these gage height data

- The `site_no` and `station_nm` can both/either serve as joining attributes
- Construct this join so that the result only includes records features where both tables have data (N=136)

17. Show the column names of this resulting spatial dataframe

18. Show the dimensions of the resulting joined dataframe

```
#14. Read the site flow data into a data frame
nebraska_site_flow <- read.csv(here('Data/Raw/NWIS_SiteFlowData_NE_RAW.csv'),
                               stringsAsFactors = TRUE)

#15. Show the structure of the dataframe
glimpse(nebraska_site_flow)
```

```
## Rows: 152
## Columns: 4
## $ site_no      <int> 6453600, 6453620, 6461500, 6461500, 6463500, 6463720, 64655~
## $ station_nm  <fct> "Ponca Creek at Verdel, Nebr.", "Missouri River blw Ponca C~
## $ date        <fct> 2019-03-20 14:15:00, 2019-03-20 14:15:00, 2019-03-20 14:15:~
## $ gage_ht     <dbl> 12.58, 19.96, 3.61, 3.63, 11.38, 3.75, 17.42, 16.31, 4.04, ~
```

#16. Join the flow data to our NWIS gage location spatial dataframe

```
nebraska_flow_gage <- nebraska_gages_sf %>%
  left_join(nebraska_site_flow,
            by = c('site_no'))
```

#17. Show the column names in the resulting spatial dataframe

```
colnames(nebraska_flow_gage)
```

```
## [1] "site_no"          "station_nm.x"      "site_tp_cd"
## [4] "coord_acy_cd"     "dec_coord_datum_cd" "station_nm.y"
## [7] "date"             "gage_ht"           "geometry"
```

#18. Show the dimensions of this joined dataset

```
dim(nebraska_flow_gage) # Result is 9 columns, 193 rows
```

```
## [1] 193  9
```

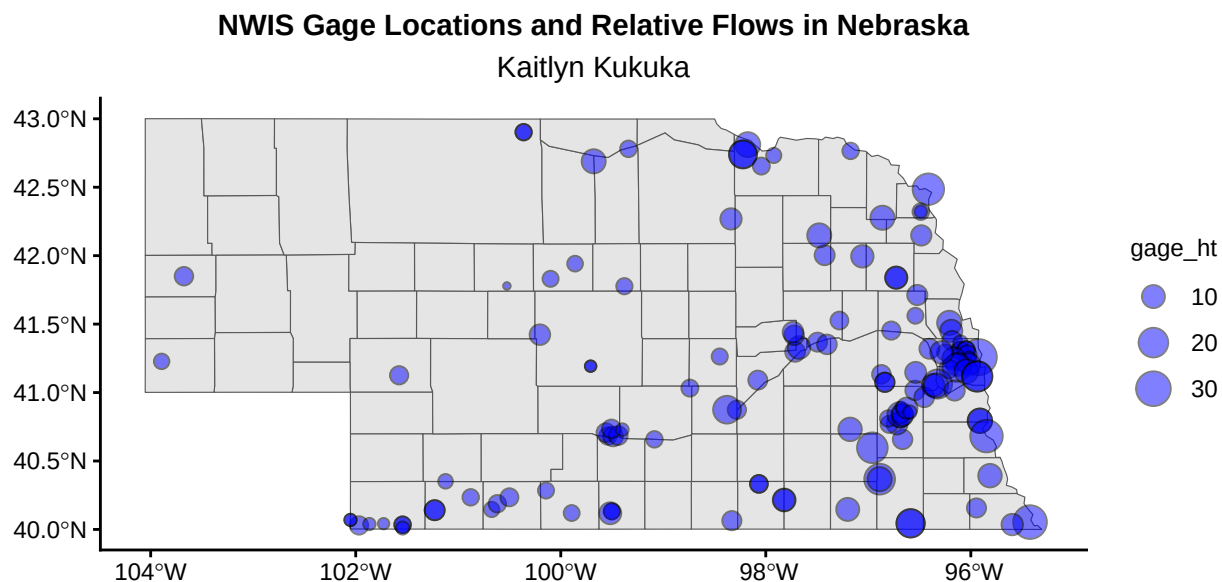
Map the pattern of gage height data

Now we can examine where the flooding appears most acute by visualizing gage heights spatially. 19. Plot the gage sites on top of counties (using `mapview`, `ggplot`, or `leaflet`) * Show the magnitude of gage height by color, shape, other visualization technique.

#Map the points, sized by gage height

```
ggplot() +
  geom_sf(data = nebraska_counties) +
  geom_sf(data = nebraska_flow_gage, aes(size = gage_ht),
          alpha = 0.5, color = 'black', fill = 'blue', shape = 21)+
  labs(title = "NWIS Gage Locations and Relative Flows in Nebraska",
        subtitle = 'Kaitlyn Kukuka')
```

```
## Warning: Removed 57 rows containing missing values or values outside the scale range
## ('geom_sf()').
```



SPATIAL ANALYSIS

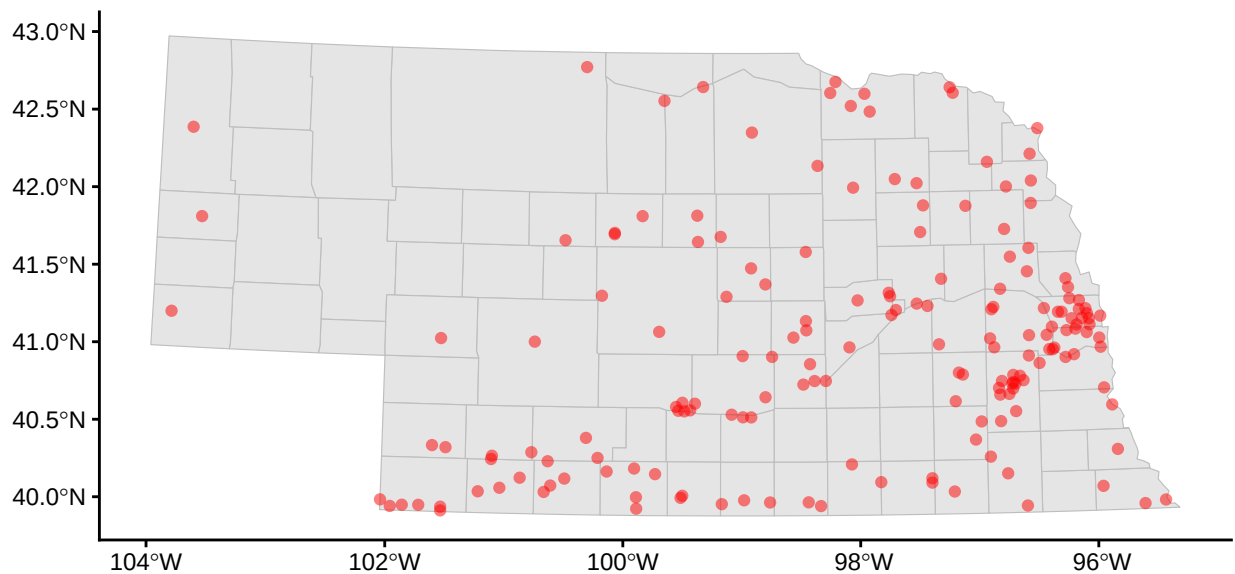
Up next we will do some spatial analysis with our data. To prepare for this, we should transform our data into a projected coordinate system. We'll choose UTM Zone 14N (EPSG = 32614).

Transform the counties and gage site datasets to UTM Zone 14N

20. Transform the counties and gage sf datasets to UTM Zone 14N (EPSG = 32614).
21. Using `ggplot`, plot the data so that each can be seen as different colors.

```
#20 Transform the counties and gage location datasets to UTM Zone 14
nebraska_counties_14N <- st_transform(nebraska_counties, 32614)
nebraska_gage_14N <- st_transform(nebraska_gages_sf, 32614)

#21 Plot the data
ggplot() +
  geom_sf(data = nebraska_counties_14N, color = "grey") +
  geom_sf(data = nebraska_gage_14N, color = 'red', alpha = 0.5)
```



Select the gages falling within a given county

Now let's zoom into a particular county and examine the gages located there. 22. Select Washington county from your projected county sf dataframe 23. Select the gage sites falling within that county to a new spatial dataframe 24. Select the gage sites within 20km of the county to a new spatial dataframe 25. Create a plot showing (each symbolized distinctly): * all Nebraska counties, * the selected county, * the gage sites in that county, * and the gage sites within 20 km of the county

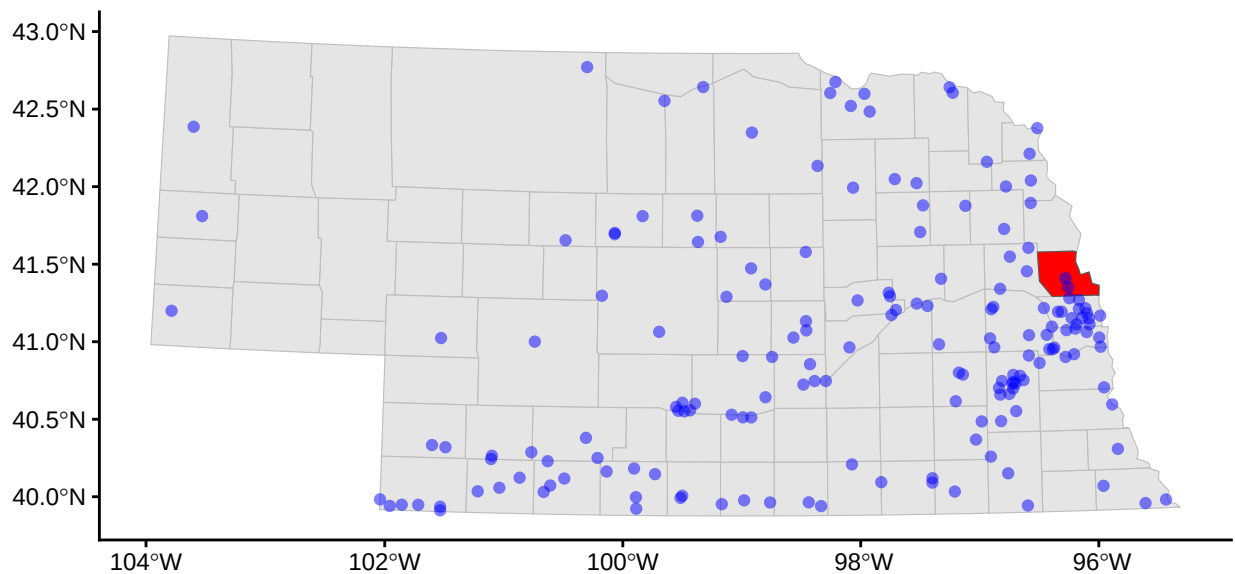
```
library(cowplot)

##
## Attaching package: 'cowplot'

## The following object is masked from 'package:lubridate':
##
## stamp

#22 Select Washington county to a new spatial dataframe
washington_county <- nebraska_counties_14N %>%
  filter(NAME == 'Washington')

ggplot() +
  geom_sf(data = nebraska_counties_14N, color = "grey") +
  geom_sf(data = washington_county, fill = 'red') +
  geom_sf(data = nebraska_gage_14N, color = 'blue', alpha = 0.5)
```



```

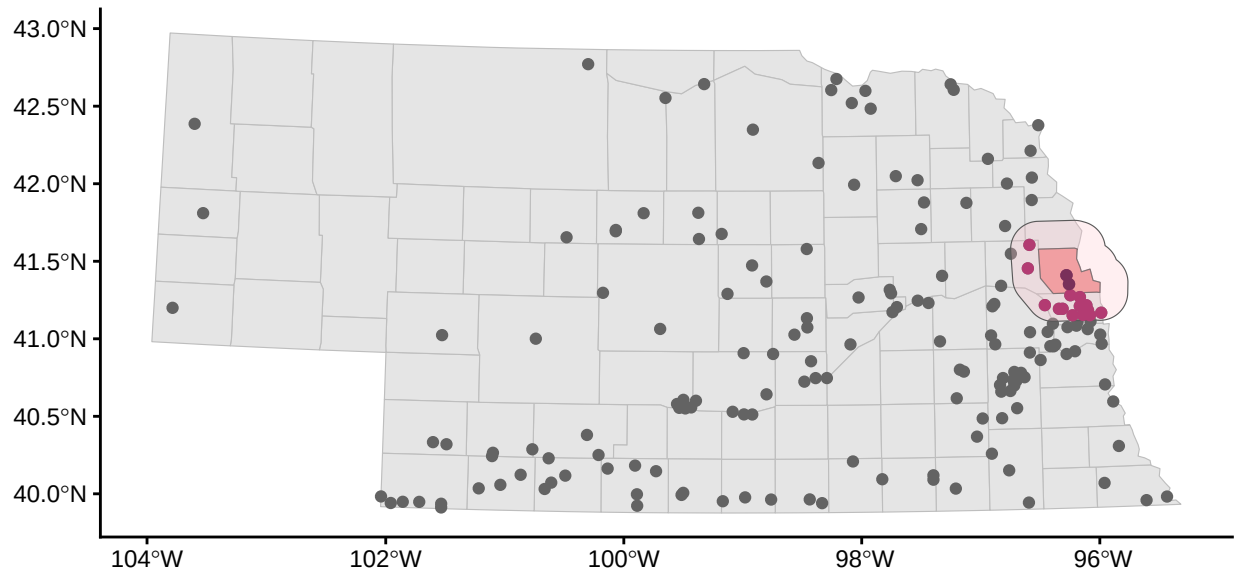
#23 Spatially select gages within the selected county to a new spatial dataframe
washington_county_gages <- st_filter(x = nebraska_gage_14N,
                                     y = washington_county)

#24 Spatially select gages within 20k of the selected county to a new spatial dataframe
washington_county_buffer <- st_buffer(x = washington_county,
                                     dist = 20000)
washington_county_gage_buffered <- st_filter(x = nebraska_gage_14N,
                                              y = washington_county_buffer)

#25 Plot
nebraska_map <- ggplot() +
  geom_sf(data = nebraska_counties_14N, color = "grey") +
  geom_sf(data = washington_county, fill = 'red', alpha = 0.35) +
  geom_sf(data = nebraska_gage_14N, color = '#636363', alpha = 1) +
  geom_sf(data = washington_county_buffer, fill = 'pink', alpha = 0.25) +
  geom_sf(data = washington_county_gage_buffered, color = '#b33b72') +
  geom_sf(data = washington_county_gages, color = '#79305a') +
  labs(title = 'Gage Stations in Nebraska, and In and Around Washington County')
nebraska_map

```

Gage Stations in Nebraska, and In and Around Washington County



```
wa_county_buffered_map <- nebraska_map +
  coord_sf(
    xlim = c(692673.1, 777040.6),
    ylim = c(4565848, 4638516)) +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.25)) +
  labs(title = 'Gage Stations within 20k km of Washington County, NE')
wa_county_buffered_map
```

Gage Stations within 20k km of Washington County, NE

